

Ethan Buckley

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Natural Sciences undergraduate (Physics major) at UCL and data science intern. Hierarchical Bayesian models fitted with MCMC, production pipelines for generated teaching materials, and the tests behind both. Seeking data science and quantitative research internships.

Education

University College London, MSci Natural Sciences

Sep 2024 – Jun 2028

Physics major, Physical Chemistry minor. Modules include Programming for Scientists, Mathematical Methods III, Quantum Physics, Atomic and Molecular Physics, Thermal Physics and Chemical Dynamics.

Experience

Data Science & Research Intern, Down Syndrome Education International

Jun 2026 – Present

- Build and extend hierarchical Bayesian models (PyMC, MCMC) of vocabulary growth in children with Down syndrome: Beta-Binomial likelihoods with age-varying dispersion and Hilbert-space Gaussian-process trajectories, fitted to checklists pooled from fourteen datasets in six countries plus around 14,800 typically-developing children from Wordbank.
- Wrote the report's data chapter: the fourteen sources, and how counts from checklists of different lengths are scored against the 810-word inventory.
- Wrote the intention-to-treat, difference-in-differences, mediation and mechanism models for a Bayesian re-analysis of a randomised reading trial (54 children, four waves), inside a causal DAG with LightGBM/SHAP predictor ranking: 124 merged pull requests, June to August 2026.
- Rebuilt a single-notebook data pipeline into per-source scripts and moved a hard-coded anonymisation key into environment secrets; Python, DuckDB, Quarto, Azure Blob Storage and Git/GitHub review.
- Build the pipeline that generates DSE's teaching images, audio and video from a written specification, and the audit, repair and upload route that files them as draft assets in the charity's catalogue. Its main rule is contrast, for the reduced contrast sensitivity common in Down syndrome.
- Chose the voices by measurement, not preference: a blind A/B arena with Bradley-Terry ranking and bootstrap intervals over four commercial text-to-speech providers, then pace calibration. 3,407 curriculum items in four accents; 7,351 images in the first production set; image unit cost halved by batch inference.

UCL Centre for Computational Science, Literature Review

From Oct 2026

- Literature review on quantum-GPU computing and variational quantum algorithms for biomolecular simulation, supervised by Professor Peter Coveney FRS. Begins October 2026.

Facilities Team Member, Roadchef, Killington Lake

Jul 2025 – Apr 2026

- Facilities team for a motorway services: site cleaning and maintenance, taking in and distributing deliveries for every outlet, and regular cover at Costa, WHSmith and Days Inn housekeeping.

Nuclear industry placements, BAE Systems (Barrow) & EDF Energy (Heysham 1)

Jul–Aug 2023

- Placements across engineering, chemistry and maintenance in safety-critical nuclear environments, including reactor-chemistry monitoring in the Heysham laboratory.

Technical skills

Programming: Python (NumPy, pandas, PyMC, LightGBM, SHAP, Matplotlib, scikit-learn, pytest), SQL / DuckDB, Git / GitHub, LaTeX, Quarto, generative-model APIs (batch inference).

Statistics & ML: Bayesian inference and MCMC, hierarchical / random-effects models, Gaussian processes, gradient boosting, causal inference (DAGs), blind A/B evaluation with Bradley-Terry ranking and bootstrap intervals, hypothesis testing, error analysis.

Projects

- Algorithmic S&P 500 stock screener (independent): combined an XGBoost classifier with a FinBERT sentiment model, trained on around 1.3 million daily observations. Built 22 technical indicators (RSI, MACD, VWAP, ATR) and used triple-barrier labelling to avoid look-ahead bias; validated with expanding-window walk-forward testing over 1,630 out-of-sample days (2020–2026): top-15 precision around 48% versus a 29% base rate.
- SEIQR epidemic cellular automaton (extended from a UCL group project): vectorised the simulation with SciPy convolution and NumPy masking for a roughly 40× speed-up, and validated it against its analytical mean-field SEIQR ODE limit. Reproducible from fixed seeds, with bit-identical regression tests in CI on Python 3.10–3.12.
- The Computable World (independent): a referenced portfolio thesis on AI, virtual twins and robotics, published as a website and PDF at ethanbuckley.github.io/computable-world.

Additional

UCL Men's Hockey Club and volunteer junior coach at Kirkby Lonsdale HC. Societies: AI, Natural Sciences. Interests: mountaineering, skiing, running.